

题目:

1. Find the residue of the following function at a specified point z_0 :

1. 求下列函数在指定点 z_0 处的留数:

(1) $\frac{1}{z-1} \exp(z^2), z_0 = 1;$

(2) $\frac{1}{(z-1)^2} \exp(z^2), z_0 = 1;$

(3) $\left(\frac{z}{1-\cos z}\right)^2, z_0 = 0;$

(4) $\frac{1}{z^2 \sin z}, z_0 = 0;$

(5) $\frac{e^z}{(z^2-1)^2}, z_0 = 1;$

2. Find the residue of the following function at each isolated singular point in the complex plane $\mathbb{C}$:

2. 求下列函数在复平面 $\mathbb{C}$ 内每一个孤立奇点处的留数:

(1) $\frac{1}{z^3 - z^5};$

(2) $\frac{1}{(1+z^2)^{m+1}}, m$ 为正整数;

(3) $\frac{z}{1-\cos z};$

(4) $\frac{\sqrt{z}}{\sinh \sqrt{z}};$

(5) $\exp\left[\frac{1}{2}\left(z - \frac{1}{z}\right)\right];$

(6) $\cos \frac{1}{\sqrt{z}};$

5. Calculate the following integrals:

5. 计算下列积分:

(1) $\int_0^{2\pi} \cos^{2n} \theta d\theta, n$ 为正整数;

(2) $\int_0^{2\pi} \frac{dx}{(a+b \cos x)^2}, a > b > 0;$

(3) $\int_0^\pi \frac{d\theta}{1+\sin^2 \theta};$

(4) $\int_0^\pi \frac{d\theta}{(1+\sin^2 \theta)^2}.$

答案:

1. 求下列函数在指定点 z_0 处的留数:

(1) $\frac{1}{z-1} \exp(z^2), z_0 = 1;$

(2) $\frac{1}{(z-1)^2} \exp(z^2), z_0 = 1;$

(3) $\left(\frac{z}{1-\cos z}\right)^2, z_0 = 0;$

(4) $\frac{1}{z^2 \sin z}, z_0 = 0;$

(5) $\frac{e^z}{(z^2-1)^2}, z_0 = 1;$

Problem 1

求下列函数在指定点 z_0 处的留数

$$(1) \frac{1}{z-1} \exp(z^2), z_0 = 1;$$

$$\operatorname{res} f(1) = \lim_{z \rightarrow 1} (z-1) f(z) = e$$

$$(2) \frac{1}{(z-1)^2} \exp(z^2), z_0 = 1;$$

$$\operatorname{res} f(1) = \lim_{z \rightarrow 1} \frac{d}{dz} [(z-1)^2 f(z)] = 2e$$

$$(3) \left(\frac{z}{1-\cos z} \right)^2, z_0 = 0;$$

$$1 - \cos z = 1 - \left(1 - \frac{1}{2!} z^2 + \frac{1}{4!} z^4 - \frac{1}{6!} z^6 + \cdots \right) = \frac{1}{2} z^2 \left(1 - \frac{1}{12} z^2 + \frac{1}{360} z^4 - \cdots \right),$$

$$\begin{aligned} \left(\frac{z}{1-\cos z} \right)^2 &= 4z^{-2} \left(1 - \frac{1}{12} z^2 + \frac{1}{360} z^4 - \cdots \right)^{-2} = 4z^{-2} \left[1 + 2 \left(\frac{1}{12} z^2 - \frac{1}{360} z^4 + \cdots \right) + O(z^4) \right] \\ &= 4z^{-2} + \frac{2}{3} + \frac{1}{45} z^2 + \cdots \end{aligned}$$

所以 $\operatorname{res} f(0) = 0$ 。

$$(4) \frac{1}{z^2 \sin z}, z_0 = 0;$$

显然 0 是三阶极点。 $\operatorname{res} f(0) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z^3 f(z) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{z(1 + \cos^2 z) - \sin 2z}{\sin^3 z}$

$$= \frac{1}{2} \lim_{z \rightarrow 0} \frac{3 \sin^2 z - 2z \sin z \cos z}{3 \sin^2 z \cos z} = \frac{1}{2} \lim_{z \rightarrow 0} \left(\frac{1}{\cos z} - \frac{2z}{3 \sin z} \right) = \frac{1}{6}$$

$$(5) \frac{e^z}{(z^2-1)^2}, z_0 = 1;$$

$$\operatorname{res} f(1) = \lim_{z \rightarrow 1} \frac{d}{dz} (z-1)^2 f(z) = 0。$$

2. 求下列函数在复平面 $\mathbb{C}$ 内每一个孤立奇点处的留数:

- | | |
|---|--|
| (1) $\frac{1}{z^3 - z^5};$ | (2) $\frac{1}{(1 + z^2)^{m+1}}, m$ 为正整数; |
| (3) $\frac{z}{1 - \cos z};$ | (4) $\frac{\sqrt{z}}{\sinh \sqrt{z}};$ |
| (5) $\exp \left[\frac{1}{2} \left(z - \frac{1}{z} \right) \right];$ | (6) $\cos \frac{1}{\sqrt{z}};$ |

Problem 2

求下列函数在复平面 $\mathbb{C}$ 内每一个孤立奇点处的留数:

(1) $\frac{1}{z^3 - z^5};$

$f(z) = \frac{1}{z^3(z+1)(z-1)}$, 0 是三阶极点, ± 1 是一阶极点。 $f(1/t) = \frac{t^5}{t^2-1}$, ∞ 点是可去奇点。

$\text{res } f(0) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z^3 f(z) = 1$, $\text{res } f(1) = \lim_{z \rightarrow 1} (z-1) f(z) = \frac{1}{2}$,

$\text{res } f(-1) = \lim_{z \rightarrow -1} (z+1) f(z) = \frac{1}{2}$ 。

(2) $\frac{1}{(1 + z^2)^{m+1}}, m$ 为正整数;

$\pm i$ 是 $m+1$ 阶极点, ∞ 点是可去奇点。

$\text{res } f(i) = \frac{1}{m!} \lim_{z \rightarrow i} \frac{d^m}{dz^m} \frac{1}{(z+i)^{m+1}} = \frac{1}{m!} \lim_{z \rightarrow i} \frac{(-1)^m (m+1)(m+2) \cdots (2m)}{(z+i)^{2m+1}}$

$= \frac{1}{(m!)^2} \frac{(-1)^m (2m)!}{(2i)^{2m+1}} = -i \frac{(2m)!}{(m!)^2 2^{2m+1}}$

$\text{res } f(-i) = i \frac{(2m)!}{(m!)^2 2^{2m+1}}$

(3) $\frac{z}{1 - \cos z};$

可看出 0 是一阶极点, $2n\pi$ ($n = \pm 1, \pm 2, \dots$) 是二阶极点, ∞ 点是非孤立奇点。

$$\operatorname{res} f(0) = \lim_{z \rightarrow 0} \frac{z^2}{1 - \cos z} = \lim_{z \rightarrow 0} \frac{z^2}{2 \sin^2 \frac{z}{2}} = 2。$$

$$1 - \cos z = 1 - \cos(z - 2n\pi) = \frac{1}{2}(z - 2n\pi)^2 \left[1 - \frac{1}{12}(z - 2n\pi)^2 + \frac{1}{360}(z - 2n\pi)^4 - + \cdots \right]$$

$$\frac{z}{1 - \cos z} = (z - 2n\pi) [1 - \cos(z - 2n\pi)]^{-1} + 2n\pi [1 - \cos(z - 2n\pi)]^{-1}$$

$$= 2(z - 2n\pi)^{-1} \left[1 + \frac{1}{12}(z - 2n\pi)^2 + O(z - 2n\pi)^4 \right]$$

$$+ 4n\pi(z - 2n\pi)^{-2} \left[1 + \frac{1}{12}(z - 2n\pi)^2 + O(z - 2n\pi)^4 \right]$$

$$= 4n\pi(z - 2n\pi)^{-2} + 2(z - 2n\pi)^{-1} + \frac{n\pi}{3} + \frac{1}{6}(z - 2n\pi) + \cdots$$

$$\text{即 } \operatorname{res} f(2n\pi) = 2。$$

$$(4) \frac{\sqrt{z}}{\sinh \sqrt{z}};$$

$$f(z) = \frac{2\sqrt{z}}{e^{\sqrt{z}} - e^{-\sqrt{z}}}, \text{ 奇点是 } -k^2\pi^2 \ (k = 0, \pm 1, \pm 2, \cdots), \ \infty \text{ 点是非孤立奇点。}$$

$$\lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \frac{2}{e^{\sqrt{z}} + e^{-\sqrt{z}}} = 1, \text{ 所以 } 0 \text{ 是可去奇点, } \operatorname{res} f(0) = 0。k \neq 0 \text{ 时,}$$

$$\operatorname{res} f(-k^2\pi^2) = \lim_{z \rightarrow -k^2\pi^2} \frac{2(z + k^2\pi^2)\sqrt{z}}{e^{\sqrt{z}} - e^{-\sqrt{z}}} = \lim_{z \rightarrow -k^2\pi^2} \frac{4z + 2(z + k^2\pi^2)}{e^{\sqrt{z}} + e^{-\sqrt{z}}} = (-1)^{k+1} 2k^2\pi^2。$$

$$(5) \exp \left[\frac{1}{2} \left(z - \frac{1}{z} \right) \right];$$

$$\operatorname{res} f(0) = -J_1(1), \operatorname{res} f(\infty) = J_1(1);$$

$$(6) \cos \frac{1}{\sqrt{z}};$$

$$f(z) = 1 - \frac{1}{2!} \left(\sqrt{\frac{1}{z}} \right)^2 + \frac{1}{4!} \left(\sqrt{\frac{1}{z}} \right)^4 - + \cdots = 1 - \frac{1}{2} z^{-1} + \frac{1}{24} z^{-2} - + \cdots$$

$$\text{即 } \operatorname{res} f(0) = -\frac{1}{2}, \quad \operatorname{res} f(\infty) = -\operatorname{res} f(0) = \frac{1}{2}.$$

5. 计算下列积分:

$$(1) \int_0^{2\pi} \cos^{2n} \theta d\theta, n \text{ 为正整数};$$

$$(2) \int_0^{2\pi} \frac{dx}{(a+b \cos x)^2}, a > b > 0;$$

$$(3) \int_0^{\pi} \frac{d\theta}{1+\sin^2 \theta};$$

$$(4) \int_0^{\pi} \frac{d\theta}{(1+\sin^2 \theta)^2}.$$

Problem 5

计算下列积分:

$$(1) \int_0^{2\pi} \cos^{2n} \theta d\theta, n \text{ 为正整数};$$

解 令 $z = e^{i\theta}$, 则 $\cos \theta = \frac{z+z^{-1}}{2}$, $d\theta = \frac{dz}{iz}$, 代入得

$$\begin{aligned} I &= \oint_{|z|=1} \left(\frac{z+z^{-1}}{2} \right)^{2n} \frac{dz}{iz} = \frac{1}{2^{2n}i} \oint_{|z|=1} \frac{(z^2+1)^{2n}}{z^{2n+1}} dz \\ &= \frac{1}{2^{2n}i} \cdot 2\pi i \cdot \frac{1}{(2n)!} \lim_{z \rightarrow 0} \frac{d^{2n}}{dz^{2n}} [(z^2+1)^{2n}] \\ &= \frac{\pi}{2^{2n-1}(2n)!} \lim_{z \rightarrow 0} \frac{d^{2n}}{dz^{2n}} \left[\sum_{k=0}^{2n} C_{2n}^k (z^2)^{2n-k} \right] \\ &= \frac{\pi}{2^{2n-1}(2n)!} \cdot (2n)! C_{2n}^n = \frac{\pi}{2^{2n-1}} C_{2n}^n. \end{aligned}$$

若 $R(\cos \theta, \sin \theta)$ 为 θ 的偶函数, 则 $\int_0^{\pi} R(\cos \theta, \sin \theta) d\theta$ 型积分亦可由上述方法求解, 因为 $\int_0^{\pi} R(\cos \theta, \sin \theta) d\theta = \frac{1}{2} \int_{-\pi}^{\pi} R(\cos \theta, \sin \theta) d\theta$.

$$(2) \int_0^{2\pi} \frac{dx}{(a+b \cos x)^2}, a > b > 0;$$

$$\text{令 } f(z) = \frac{1}{z \left(a + b \frac{z^2+1}{2z} \right)^2} = \frac{4}{b^2} \frac{z}{(z-z_1)^2 (z-z_2)^2}, \text{ 其中 } z_1 = -\frac{a}{b} - \sqrt{\left(\frac{a}{b} \right)^2 - 1},$$

$$z_2 = -\frac{a}{b} + \sqrt{\left(\frac{a}{b}\right)^2 - 1}, \quad -\frac{a}{b} - \sqrt{\left(\frac{a}{b}\right)^2 - 1} < -\frac{a}{b} < -1, \text{ 所以 } |z_1| > 1, \text{ 在单位圆外,}$$

$$|z_2| = \left| -\frac{a}{b} + \sqrt{\left(\frac{a}{b}\right)^2 - 1} \right| = 1 / \left| -\frac{a}{b} - \sqrt{\left(\frac{a}{b}\right)^2 - 1} \right| < 1, \text{ 在单位圆内.}$$

$$\operatorname{res} f(z_2) = \lim_{z \rightarrow z_2} \frac{d}{dz} \left[(z - z_2)^2 f(z) \right] = \frac{a}{(a^2 - b^2)^{3/2}},$$

$$\text{所以原积分} = \frac{2\pi a}{(a^2 - b^2)^{3/2}}.$$

$$(3) \int_0^\pi \frac{d\theta}{1 + \sin^2 \theta};$$

$$\text{原积分} = \int_0^\pi \frac{2d\theta}{3 - \cos 2\theta} = \int_0^{2\pi} \frac{d\varphi}{3 - \cos \varphi}$$

$$\text{令 } f(z) = \frac{1}{z} \frac{2}{\left(3 - \frac{z^2 + 1}{2z}\right)^2} = \frac{8z}{(z - 3 - 2\sqrt{2})^2 (z - 3 + 2\sqrt{2})^2}, \text{ 则}$$

$$\operatorname{res} f(3 - 2\sqrt{2}) = \lim_{z \rightarrow 3 - 2\sqrt{2}} \frac{d}{dz} \frac{8z}{(z - 3 - 2\sqrt{2})^2} = \frac{3}{8\sqrt{2}}, \text{ 原积分} = \frac{3}{4\sqrt{2}} \pi.$$

$$(4) \int_0^\pi \frac{d\theta}{(1 + \sin^2 \theta)^2}.$$

$$\text{原积分} = \int_0^\pi \frac{4d\theta}{(3 - \cos 2\theta)^2} = \int_0^{2\pi} \frac{2d\varphi}{(3 - \cos \varphi)^2}$$

$$\text{令 } f(z) = \frac{1}{z} \frac{2}{\left(3 - \frac{z^2 + 1}{2z}\right)^2} = \frac{8z}{(z - 3 - 2\sqrt{2})^2 (z - 3 + 2\sqrt{2})^2}, \text{ 则}$$

$$\operatorname{res} f(3 - 2\sqrt{2}) = \lim_{z \rightarrow 3 - 2\sqrt{2}} \frac{d}{dz} \frac{8z}{(z - 3 - 2\sqrt{2})^2} = \frac{3}{8\sqrt{2}}, \text{ 原积分} = \frac{3}{4\sqrt{2}} \pi$$